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$$2(3^x + 2^{x-1}) = 3(2^x - 3^{x-1})$$

$$\Leftrightarrow 2 \cdot 3^x + 2^x = 3 \cdot 2^x - 3^x$$

$$\Leftrightarrow 2 \cdot 3^x + 2^x - 3 \cdot 2^x + 3^x = 0$$

$$\Leftrightarrow 3^x(2 + 1) + 2^x(-3 + 1) = 0$$

$$\Leftrightarrow 3^{x+1} \cdot (-2^{x+1}) = 0$$

$$\Leftrightarrow 3^{x+1} = 2^{x+1}(\text{delen door: } 2^x)$$

$$\Leftrightarrow \frac{3^{x+1}}{2^x} = 2$$

$$\Leftrightarrow 3 \cdot \left(\frac{3}{2}\right)^x = 2$$

$$\Leftrightarrow \left(\frac{3}{2}\right)^x = \frac{2}{3}$$

$$\Leftrightarrow x = -1$$

References